(Start writing below the box)

Linear differential equation:
A differential equation of the form $\frac{dy}{dx} + p(x)y = Q(x)$ is called 1st order linear differential equation.

Working procedure:

(I) Make the DE of the form $\frac{dy}{dx} + p(x)y = Q(x)$

(II) find the I.F by $I.F = e^{\int P dx}$

(III) Obtain the solution by multiplying the equation by I.F.

Question 1: Solve $\frac{dy}{dx} + xy = x$

Soln: Given that $\frac{dy}{dx} + xy = x \quad \cdots (1)$

Comparing the equation with standard form

$$p = x \text{ and } Q = x$$

The integrating factor,

$$I.F = e^{\int P dx} = e^{\int x dx} = e^{x^2/2}$$

Multiplying eqn(1) by $e^{x^2/2}$ we get,

$$e^{x^2/2}\frac{dy}{dx} + xye^{x^2/2} = xe^{x^2/2}$$

$$\Rightarrow e^{x^2/2}dy + xye^{x^2/2}dx = xe^{x^2/2}dx$$

By integrating we get,

$$\int e^{x^2/2}dy + \int xye^{x^2/2}dx = \int xe^{x^2/2}dx$$

$$\Rightarrow ye^{x^2/2} +$$

The solution is,

$$y\cdot e^{\int Pdx} = \int Q\cdot e^{\int Pdx}dx$$

$$\Rightarrow y\cdot e^{x^2/2} = \int x\cdot e^{x^2/2}dx$$

$$\Rightarrow y\cdot e^{x^2/2} = \int e^{x^2/2}\, d(x^2/2)$$

$$\Rightarrow ye^{x^2/2} = e^{x^2/2}+c$$

$$\Rightarrow y = e^{-x^2/2}e^{x^2/2}+ce^{-x^2/2}$$

$$\therefore y = 1+ce^{-x^2/2}$$ Ans

Question 02 Solve $(1+y^2)dx = (\tan^{-1}y - x)dy$

Solution: Here, $(1+y^2)dx = (\tan^{-1}y - x)dy$

$$\Rightarrow \frac{dx}{dy} = \frac{\tan^{-1}y - x}{1+y^2}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\tan^{-1}y}{1+y^2} - \frac{x}{1+y^2}$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\tan^{-1}y}{1+y^2}$$

Here, $P = \frac{1}{1+y^2}$ and $Q = \frac{\tan^{-1}y}{1+y^2}$

$$I.F = e^{\int P\,dy} = e^{\int \frac{1}{1+y^2}dy} = e^{\tan^{-1}y}$$

The solution is $x \cdot e^{\int P dy} = \int Q \cdot e^{\int P dy}\,dy$

$$\Rightarrow x e^{\tan^{-1}y} = \int \frac{\tan^{-1}y}{1+y^2} e^{\tan^{-1}y}\,dy$$

$$\Rightarrow x e^{\tan^{-1}y} = \int \tan^{-1}y \cdot e^{\tan^{-1}y}\, d(\tan^{-1}y)$$

$$\Rightarrow x e^{\tan^{-1}y} = \tan^{-1}y\, e^{\tan^{-1}y} - e^{\tan^{-1}y} + c$$

$$\Rightarrow x = e^{-\tan^{-1}y} \tan^{-1}y\, e^{\tan^{-1}y} - e^{-\tan^{-1}y} e^{\tan^{-1}y} + c e^{-\tan^{-1}y}$$

$$\therefore x = \tan^{-1}y - 1 + c e^{-\tan^{-1}y}.$$

Ans

Ex Solve $(1+x^2)\frac{dy}{dx} + 2xy = 4x^2$

Soln: Given that $(1+x^2)\frac{dy}{dx} + 2xy = 4x^2$

$\Rightarrow \frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{4x^2}{1+x^2}$

Here $P = \frac{2x}{1+x^2}$ and $Q = \frac{4x^2}{1+x^2}$

$I.F = e^{\int P dx} = e^{\int \frac{2x}{1+x^2}dx} = e^{\ln(1+x^2)} = (1+x^2)$

The solution is, $y \cdot e^{\int P dx} = \int Q \cdot e^{\int P dx} dx$

$y(1+x^2) = \int \frac{4x^2}{1+x^2}(1+x^2) dx$

$y(1+x^2) = \int 4x^2 dx$

$y(1+x^2) = 4\frac{x^3}{3} + c$

$3y(1+x^2) = 4x^3 + c$ Ans

Ex Solve $\frac{dy}{dx} + y\tan x - \sec x = 0$

Soln: Given that $\frac{dy}{dx} + y\tan x - \sec x = 0$

$\frac{dy}{dx} + y\tan x = \sec x$

Here, $P = \tan x$ and $Q = \sec x$

Now. $I.F = e^{\int \tan x dx} = e^{\ln \sec x} = \sec x$

The soln is $y \sec x = \int \sec x \cdot \sec x dx$

$\Rightarrow y \sec x = \int \sec^2 x\,dx$

$\Rightarrow y \sec x = \tan x + c$ Ans

Ex Solve $(1-x^2)\frac{dy}{dx} - xy = 1$

Soln: Given that, $(1-x^2)\frac{dy}{dx} - xy = 1$

$$\Rightarrow \frac{dy}{dx} - \frac{xy}{(1-x^2)} = \frac{1}{1-x^2}$$

Here, $P = \frac{-x}{1-x^2}$ and $Q = \frac{1}{1-x^2}$

$$I.F = e^{\int \frac{-x}{1-x^2}dx} = e^{\int \frac{-2x}{2(1-x^2)}dx} = e^{\frac{1}{2}\ln(1-x^2)}$$

$$= e^{\ln(1-x^2)^{\frac{1}{2}}} = (1-x^2)^{\frac{1}{2}} = \sqrt{1-x^2}$$

The solution is

$$y\cdot\sqrt{1-x^2} = \int \frac{1}{1-x^2}\sqrt{1-x^2}\,dx$$

$$y\cdot\sqrt{1-x^2} = \int \frac{1}{\sqrt{1-x^2}}\,dx$$

$$y\cdot\sqrt{1-x^2} = \sin^{-1}x + c$$ Ans

Ex Solve $\frac{dy}{dx} + y\cot x = 5e^{\cos x}$

Soln: $\frac{dy}{dx} + \cot x\, y = 5e^{\cos x}$

Here, $P = \cot x$, $Q = 5e^{\cos x}$

$$I.F = e^{\int P dx} = e^{\int \cot x\,dx} = e^{\ln \sin x} = \sin x$$

The general solution is,

$y \sin x = \int 5e^{\cos x} \cdot \sin x \, dx$

$y \sin x = -5\int dz$

$y \sin x = -5z + c$

$y \sin x = -5e^{\cos x} + c$ Ans

let,

$e^{\cos x} = z$

$e^{\cos x}(-\sin x) dx = dz$

$\Rightarrow e^{\cos x} \sin x \, dx = -dz$

Bernoulli equation: The DE of the form

$\frac{dy}{dx} + P(x)y = Q(x)y^n$ is called bernoulli differential equation.

Working procedure:

(i) Make the DE in the form $\frac{dy}{dx} + Py = Q$

(ii) Find the $I.F = e^{\int P dx}$

(iii) Find the g.s by $y \cdot e^{\int P dx} = \int Q e^{\int P dx} + c$

Solve $\frac{dy}{dx} + y = y^2 e^x$

Soln

$\frac{dy}{dx} + y = y^2 e^x$

$\Rightarrow \frac{-1}{y^2}\frac{dy}{dx} + \frac{-1}{y} = -e^x$ [Divided by $-y^2$]

$\Rightarrow \frac{d\left(\frac{1}{y}\right)}{dx} - \frac{1}{y} = -e^x$ [$\because dy^m = my^{m-1}dy$]

$\Rightarrow$

$\therefore I.F = e^{\int -1 dx} = e^{-x}$

$\therefore$ The general soln is

$$\frac{1}{y} e^{\int P dx} = \int Q e^{\int P dx}$$

$$\Rightarrow \frac{1}{y} e^{-x} = \int -e^{x} \cdot e^{-x} dx$$

$$\Rightarrow \frac{1}{y} e^{-x} = -\int dx$$

$$\Rightarrow \frac{1}{y} e^{-x} = -x + c$$

$$\Rightarrow e^{-x} = y(c-x) \text{ Ans}$$

Solve $\frac{dy}{dx} + \frac{1}{x} \sin 2y = x^3 \cos^2 y$

Soln: Given that $\frac{dy}{dx} + \frac{1}{x} \sin 2y = x^3 \cos^2 y$

$$\Rightarrow \frac{1}{\cos^2 y} \frac{dy}{dx} + \frac{1}{x} \frac{\sin 2y}{\cos^2 y} = x^3 \quad [\text{Divided by } \cos^2 y]$$

$$\Rightarrow \sec^2 y \frac{dy}{dx} + \frac{1}{x} \frac{2 \sin y \cos y}{\cos^2 y} = x^3$$

$$\Rightarrow \sec^2 y \frac{dy}{dx} + \frac{2}{x} \tan y = x^3$$

$$\Rightarrow \frac{d(\tan y)}{dx} + \frac{2}{x} \tan y = x^3$$

Here, $I.F = e^{\int P dx} = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = e^{\ln x^2} = x^2$

The general solution is:

$$(\tan y) x^2 = \int x^3 \cdot x^2 dx$$

$$\tan y \; x^2 = \frac{x^6}{6} + c$$

$$x^2 \tan y = \frac{x^6}{6} + c \quad \text{Ans}$$

5. Solve $2\frac{dy}{dx} = \frac{y}{x} + \frac{y^2}{x^2}$

__Soln:__ Given that $2\frac{dy}{dx} = \frac{y}{x} + \frac{y^2}{x^2}$

$$\Rightarrow 2\frac{dy}{dx} - \frac{y}{x} = \frac{y^2}{x^2}$$

$$\Rightarrow \frac{1}{y^2}\frac{dy}{dx} - \frac{1}{2xy} = \frac{1}{2x^2} \quad [\text{Divided by } 2y^2]$$

$$\Rightarrow -\frac{d\left(\frac{1}{y}\right)}{dx} - \frac{1}{2xy} = \frac{1}{2x^2}$$

$$\Rightarrow \frac{d\left(\frac{1}{y}\right)}{dx} + \frac{1}{2x}\cdot\frac{1}{y} = -\frac{1}{2x^2}$$

Here $I\cdot F = e^{\int \frac{1}{2x}dx} = e^{\frac{1}{2}\ln x} = e^{\ln x^{1/2}} = \sqrt{x}$

The general solution is: $\frac{1}{y}\sqrt{x} = \int -\frac{1}{2x^2}\sqrt{x}\,dx$

$$\Rightarrow \frac{1}{y}\sqrt{x} = -\frac{1}{2}\int x^{-3/2}dx$$

$$\Rightarrow \frac{1}{y}\sqrt{x} = -\frac{1}{2}\frac{x^{-1/2}}{-1/2} + c$$

$$\Rightarrow \frac{1}{y}\sqrt{x} = \frac{1}{\sqrt{x}} + c$$

$$\Rightarrow \sqrt{x} = y\left(\frac{1}{\sqrt{x}} + c\right) \text{ Ans.}$$